GRADE 9 • DIGITAL TECHNOLOGY & INNOVATION

NUMBER SYSTEM MADE EASY

1 0 1 1 0 1

0 1 0 1 1 0

1 1 0 0 1 0

VIDEO 03

BINARY → DECIMAL

Using the Positional Method

From bits to everyday numbers — step by step.

What will we learn today?

By the end of this lesson, you should be able to convert a binary number into its decimal equivalent.

01

Recognise the bases

Binary is base 2; decimal is base 10.

02

Use powers of 2

Assign positional values from right to left.

03

Calculate the value

Multiply each bit by its position and add.

First, remember the two systems

Our task is to express a number written in binary using decimal digits.

BINARY

BASE 2

Uses only 0 and 1

DECIMAL

BASE 10

Uses digits 0 through 9

Today: 101101_2 → $?_{10}$

Why do we use powers of 2?

Binary is a base-2 number system, so each position represents a power of 2.

2^0	2^1	2^2	2^3	2^4	2^5	2^6
1	2	4	8	16	32	64

Start at the RIGHT with 2^0, then increase the power as you move LEFT.

$2^0 = 1$ • $2^1 = 2$ • $2^2 = 4$ • $2^3 = 8$

Place the powers underneath the bits

Example binary number: 101101_2

1	0	1	1	0	1
2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1

Six binary digits → six positional values

Each digit is matched with the power directly below it.

Convert 101101_2 into decimal

Step 1: Multiply each binary digit by its positional value.

1×2^5	$= 32$
0×2^4	$= 0$
1×2^3	$= 8$
1×2^2	$= 4$
0×2^1	$= 0$
1×2^0	$= 1$

A zero contributes 0; a one contributes the full positional value.

Step 2: Add the results

Now add all the values obtained in the previous step.

$$32 + 0 + 8 + 4 + 0 + 1$$

$$45$$

FINAL ANSWER

$$101101_2 = 45_{10}$$

Only add the positions where the bit is 1

You can skip calculating the zero positions because zero multiplied by any value is zero.

1	0	1	1	0	1
32	16	8	4	2	1
32	—	8	4	—	1

32 + 8 + 4 + 1 = 45

The 0-bits add nothing, so leave those values out.

Convert 11001_2 into decimal

There are five digits, so the positional values are 16, 8, 4, 2 and 1.

Binary digit	1	1	0	0	1
Power of 2	2^4	2^3	2^2	2^1	2^0
Value	16	8	4	2	1
Product	16	8	0	0	1

$$16 + 8 + 0 + 0 + 1 = 25$$

$$11001_2 = 25_{10}$$

Common mistake: starting at 2^1

The rightmost binary digit always has a positional power of zero.

INCORRECT

$$\ldots\ 2^3\ \ 2^2\ \ 2^1$$

Do not begin with 2^1 on the right.

CORRECT

$$\ldots\ 2^3\ \ 2^2\ \ 2^1\ \ 2^0$$

Start from 2^0 at the right, then count up.

RIGHT → LEFT: $2^0, 2^1, 2^2, 2^3, \ldots$

Binary → Decimal in four steps

Follow this sequence every time you solve a conversion question.

1

Write

Write the binary number.

→

2

Assign

Assign powers of 2 from right to left.

→

3

Multiply

Multiply each digit by its positional value.

→

4

Add

Add all the results to get the decimal value.

Your turn!

Convert the following binary number into decimal.

20-SECOND CHALLENGE

$$10101_2 \rightarrow ?_{10}$$

Hint: start from the right with 2^0.

Pause the video and work it out before revealing the answer.

Let's check your answer

Multiply each bit by its positional value, then add.

1×2^4	0×2^3	1×2^2	0×2^1	1×2^0
= 16	= 0	= 4	= 0	= 1

$$16 + 4 + 1 = 21$$

$$10101_2 = 21_{10}$$

Can you solve this one independently?

Use the powers of 2: 32, 16, 8, 4, 2 and 1.

FINAL CHALLENGE • 20 SECONDS

$$111001_2 \rightarrow ?_{10}$$

Which positional values do you need to add?

Pause the video and try it before the answer appears.

Answer: 57_{10}

For 111001_2, add the positional values where the bit is 1.

1	1	1	0	0	1
32	16	8	4	2	1

$$32 + 16 + 8 + 0 + 0 + 1$$

$$32 + 16 + 8 + 1 = 57$$

$$111001_2 = 57_{10}$$

What should you remember?

Binary to decimal = positional values based on powers of 2.

BASE

Binary is base 2.

START

Begin at 2^0 on the right.

MULTIPLY

Multiply each bit by its position.

ADD

Add the results to get decimal.

Powers of 2 + multiplication + addition

GREAT WORK!

You can now convert binary numbers into decimal.

COMING UP IN VIDEO 4

DECIMAL → BINARY

Using the Division–Remainder Method

SEE YOU IN THE NEXT VIDEO!